Invoice Generator Web App – Project Report

An automated invoicing system for SN Advertising · Prepared: 2026-11-09

1. Introduction and Background

SN Advertising is an advertising and marketing agency that provides services to many different companies. Today, invoices for every client are produced by manually editing an Excel file each time. The client list is large, and each client's invoice differs depending on the tasks requested.

The goal of this report is to describe a web app that removes the manual Excel editing and automatically generates invoices based on the client and the work requested, and to set out the features, changes, data structure, and step-by-step process needed to build it.

2. Current Situation (Analysis of Sample Invoices)

Three sample invoices were reviewed:

File	Client	Description	Amount (Rs.)	VAT	Cheque payee
Araliya_Land_Video_Aug_2026	Araliya Lands and Homes (Pvt) Ltd	Video Campaign with Presenter (3 Clips)	125,000	None	SN Advertising
Chamara_Stikers_Ran_Fm_OCT_2026	Chamara Motors and Stickers Center	Sponsorship Package on Run FM (Oct–Nov 2026)	300,000 + VAT 54,000 = 354,000	Yes (18%)	HARIPARA (PVT) LTD
Chamara_Stikers_Ridam_Fm_Oct_2026	Chamara Motors and Stickers Center	Sponsorship Package on Ridam FM Evening Show (Oct–Nov 2026)	200,000	None	HARIPARA (PVT) LTD

2.1 Common Invoice Layout

- Letterhead at the top: SN Advertising address, phone numbers, email
- Advertiser (client) name and address
- Date, Invoice No, Product, Reg. No, Your Ref, VAT No
- Work description and Amount (Rs.)
- VAT (where applicable), Total, Grand Total

- Authorized Signature
- Cheque and bank transfer instructions at the bottom

2.2 Issues and Variations Observed

- **Inconsistent invoice number formats**: 23458, 23466 and 4415/RFM show numbers are typed by hand with no standard format.
- **Invoice No and Reg. No swapped**: on the Ridam FM invoice, Invoice No is 4415/RFM and Reg. No is 23465, which is the reverse of the other invoices.
- **VAT on some invoices only**: the same client (Chamara) has one invoice with VAT and one without, so VAT must be decided per invoice.
- **Cheque payee varies**: SN Advertising or HARIPARA (PVT) LTD, meaning invoices are issued under more than one billing entity.
- **Spelling errors**: "SPONSER", "PACKEGE", "Discription", "croeesd".
- **Dates stored as Excel serial numbers** (for example 46246 = 2026-08-12, 46295 = 2026-09-30).
- **Special notes typed in manually**, such as "75% Advance Payment to Proceed with Work".
- **Limited line items**: a single description line only, with no quantity × unit price structure.

3. Project Objectives

1. Stop editing an Excel file by hand for every invoice.
2. Generate an invoice in a few steps by selecting the client and the requested work.
3. Calculate VAT, totals and invoice numbers automatically and accurately.
4. Keep all clients, invoices and payments in one place.
5. Produce a PDF that keeps the current invoice layout (letterhead).

4. Changes Needed to the Current Invoice

Change	Detail
Invoice number	A fixed format (for example `SN-2026-0001` or `CHM-2026-0003`) assigned automatically by the system.
Invoice No / Reg. No	Separate, clearly defined fields so they can no longer be swapped.

Change	Detail
Spelling errors	Corrected once in the template (Sponsorship, Package, Description, crossed).
VAT	A per-invoice checkbox. The rate (currently 18%) is stored in settings so it can be changed in one place.
Cheque payee and bank details	Switch automatically based on the issuing company (SN Advertising or HARIPARA).
Date	Shown in a normal date format (YYYY-MM-DD or DD/MM/YYYY).
Description lines	Multiple line items (description, quantity, unit price, line total).
Signature	An uploaded signature image placed on the PDF automatically.
Notes	Standard notes such as "75% Advance" chosen from templates.

5. Features to Add

Phase 1 – Essential (MVP)

- User login (separate permissions for Owner and Staff)
- Client management
- Service / Task catalog with default prices
- Billing entities (SN Advertising, HARIPARA (PVT) LTD, etc.)
- Invoice generator: select client, add tasks, tick VAT, automatic number, save invoice
- PDF generation and download
- Invoice list with search and filters

Phase 2 – Important

- Payment status: Draft / Sent / Partially Paid / Paid / Overdue
- Recording advance payments and outstanding balances
- Duplicate an invoice (for recurring monthly sponsorships)
- Send the PDF by WhatsApp or email
- Excel export for the accountant

Phase 3 – Extras

- Convert a quotation into an invoice
- Generate receipts
- Dashboard (monthly revenue, unpaid amounts, reports by client)
- Overdue payment reminders
- Audit log (who changed what)
- Prevent deleting invoices; allow voiding only

6. Information the Client Form Must Capture

There are many clients, and each may have different details. Fields marked with * are mandatory.

6.1 Basic Details

Field	Note
Client / Company name *	The name printed on the invoice
Short name (display name)	For quick searching in lists
Address *	Several lines (line 1, line 2, city)
Business type	Pvt Ltd / Sole proprietor / Individual

6.2 Contact

Field	Note
Contact person name	
Phone numbers	More than one can be added
WhatsApp number	For sending invoices
Email	

6.3 Tax and Legal

Field	Note
Client VAT number	If applicable (for example Chamara: 100958996-7000)
Charge VAT?	Yes / No (default value)
TIN / Business Reg. number	If required

6.4 Default Billing Settings

Field	Note
Default billing entity	SN Advertising or HARIPARA (PVT) LTD
Payment terms	Immediate / 7 days / 30 days
Advance %	For example 75%
Default note	For example "75% Advance Payment to Proceed with Work"
Currency	LKR

6.5 Internal Use

Field	Note
Client Code	A short unique code (CHM, ARL) that can also be used in invoice numbers
Status	Active / Inactive
Internal notes	Not visible to the client
Date added	Filled in automatically

6.6 Optional

- Logo
- Default services (work usually provided to this client, with prices)
- Client bank details
- Additional addresses / branches

6.7 Important Rules

- **Snapshot**: when an invoice is created, copy the client's name, address and VAT number onto the invoice itself. Later changes to the client record must not alter old invoices.
- **Duplicate check**: warn when a client is added with a name or VAT number that already exists.

7. Data Structure (Database Tables)

Table	Main fields
Users	Name, email, password (hashed), role (Owner / Staff)
Clients	All fields in section 6
Services	Name, description, default price, VAT applicable
ClientServices	Client + service + client-specific price
BillingEntities	Name, address, phone, email, bank details, cheque payee, VAT number, invoice number prefix, signature image
Invoices	Number, client, billing entity, date, VAT flag, VAT %, subtotal, VAT amount, Grand Total, status, notes, client snapshot
InvoiceItems	Invoice, description, period (for example OCT–NOV 2026), quantity, unit price, line total
Payments	Invoice, date, amount, method (Cash / Cheque / Bank transfer), notes
Settings	VAT rate, invoice numbering scheme, default notes
AuditLog	Who, when, what changed

Relationships: One Client has many Invoices. One Invoice has many InvoiceItems and Payments. Each Invoice belongs to one BillingEntity.

8. How an Invoice Is Generated

1. Click "New Invoice".
2. Select the client; their details and default settings fill in automatically.

3. Confirm the billing entity (defaults to the client's).
4. Choose the date and the service period.
5. Add tasks from the catalog or type a new line.
6. Tick VAT if applicable. Subtotal, VAT and Grand Total calculate automatically.
7. Choose an advance % or note if needed.
8. Preview and save. The invoice number is assigned automatically.
9. Download the PDF, or send it by WhatsApp or email.

9. Technology Options

Option	Pros	Cons	Best for
Google Sheets + Apps Script	Fast, low cost, familiar like Excel	Limited as it grows, less control over PDF layout	A very small start
Web app (Next.js or Django + PostgreSQL)	Full control, good PDF output, easy to extend, secure	Longer development time	**Recommended option**
No-code (Airtable / Glide)	Quick	Hard to match the PDF layout exactly, monthly fees	A prototype

Recommendation: a web app with a PostgreSQL database and PDF generation from an HTML template, using a responsive design that works on both phones and computers.

10. Implementation Steps

Step	Activity	Estimated time
1	Confirm requirements: meet the client, get answers to the questions at the end of this report, and analyse 10–20 past invoices.	2–3 days
2	Design: database structure, screen sketches, and client approval of the PDF template.	3–5 days
3	Choose the technology and set up the environment.	1–2 days
4	Build the core: Login → Clients → Services → Billing entities → Invoice generator → PDF.	1–2 weeks

Step	Activity	Estimated time
5	Testing: create real invoices in the new system and compare totals (especially VAT) with the old Excel ones.	3–5 days
6	Deploy: server, domain, HTTPS and automatic backups.	1–2 days
7	Data import and training: import existing clients and past invoices; train staff.	2–3 days
8	Parallel run: use both Excel and the new system.	2–4 weeks
9	Add Phase 2 and Phase 3 features based on feedback.	Ongoing

11. Risks and Safeguards

- **Incorrect VAT calculation** can cause legal issues. Have an accountant approve the logic.
- **Duplicate or skipped invoice numbers**: numbers must be assigned only by the system.
- **Data loss**: daily automatic backups are mandatory.
- **Unauthorised access**: use login, role-based permissions and HTTPS.
- **Changing issued invoices**: void and re-issue instead of editing, and keep an audit log.
- **Fallback**: keep the Excel files during the parallel run so there is a way back.

12. Details to Request from the Client

The following information should be collected from the client before the plan above is finalised.

12.1 Business Rules

1. What format should invoice numbers follow (simple running number, year/month based, or client-code based)? What is the latest invoice number in use?
2. Should old invoice numbers (23458, 4415/RFM) be carried into the new system?
3. Is the VAT rate always 18%? In which cases is VAT charged (only when the client has a VAT number, or under other rules)?

4. How many billing entities are there (currently SN Advertising and HARIPARA (PVT) LTD)? For each: address, VAT number, bank account and cheque payee name. Is there a rule for which client uses which entity?
5. What exactly do "Reg. No", "Product" and "Your Ref" mean, and how should they be filled in?
6. For advance payments (for example 75%), is a separate invoice issued for the advance and for the balance, or is it only a note on one invoice?
7. Are there payment terms (number of days), late fees or discounts?
8. How are credit notes and invoice corrections handled today?

12.2 Data

9. The full client list (in Excel), with addresses, VAT numbers and contacts.
10. Previously issued invoice Excel files (at least one year) so all variations can be reviewed.
11. The types of work requested (video campaign, radio sponsorship, stickers, social media, etc.) and their typical prices.
12. Approximately how many clients and how many invoices per month?

12.3 Invoice Layout

13. Should the current letterhead layout stay exactly as is, or is a new design wanted? Is a high-quality logo available?
14. Should a signature image be included? Whose signature?
15. Should invoices be in English only, or Sinhala as well?
16. Do invoices usually have only a few line items? Is quantity × unit price needed?

12.4 Users and Access

17. How many people will use the system? Who creates invoices, and who only views them?
18. How are invoices sent to clients today (WhatsApp, email, printed)?
19. Is payment tracking (Paid / Pending) required?

12.5 Additional Needs

20. Are quotations, receipts or monthly reports needed?
21. Is a dashboard (revenue, unpaid amounts) needed?
22. Is an Excel export for the accountant needed? Is accounting software (Tally, QuickBooks, etc.) in use?

23. Will the system be used on mobile phones, computers, or both?
24. Budget and timeline: by when is the system needed? Is there a limit for hosting and monthly running costs?
25. Data privacy: are there special requirements for data ownership or where backups are stored?

13. Conclusion

The current Excel-based process is prone to errors and inconsistent formats. A web app that generates invoices automatically from the client, the service and the billing entity will improve speed, accuracy and data management. The recommended approach is to first build the MVP (Clients, Services, Billing entities, Invoice PDF generator), then add payment tracking and reporting. Once the client answers the questions in section 12, a final plan and timeline can be confirmed.